Two ternary codes from flags and Fano subplanes

Private derivation and finite certificate: review pending

This note addresses the 120-coordinate modules N and K in L050, Theorem 1(b) and Proposition 3. The source is Leemans–Rodrigues, *Linear codes with complementary duals from some strongly regular subgraphs of the McLaughlin graph*, Math. Commun. 21 (2016), 239–249, particularly printed pages 240 and 247. Its construction of the residual Steiner graph is retained. We supply an incidence operator producing the 15-dimensional code, followed by complete finite rank, module, weight and permutation-group certificates. The 21-dimensional graph code is a different subspace; its rank or LCD property does not establish the assertions here. The original uses constructive MeatAxe arguments and retained representation theory. Our discovery also used MeatAxe, but the final module certificate checks an explicit filtration and all linear intertwiner spaces instead. No historical novelty or whole-paper recovery is asserted.

1 The incidence operator

Use the fixed $S(4, 7, 23)$ already constructed and checked by `steiner_construction.py`. Delete two named points. The 120 heptads avoiding them form a set $\mathcal{H}$. The 21 heptads containing both points, with those two removed, form the lines of a projective plane on 21 points. Its lines have size five; every two remaining points lie on one line by the Steiner property. Two such lines meet in one point, since the original heptads meet in one or three points and already share the deleted pair.

For $H \in \mathcal{H}$, every two points of H lie on a plane line meeting H in three points. Indeed the heptad defining that line shares at least two points with H and hence exactly three. These triples form the Fano subplane on the seven points of H. Let $\mathcal{F}$ be the 105 incident point–line flags of the plane, and define a 105×120 matrix R by

$$R_{(p,\ell),H} = 1 \quad \Longleftrightarrow \quad p \in H \ \text{ and } \ |\ell \cap H| = 3.$$

Thus each column records the 21 flags of the corresponding Fano subplane. All following code spaces and matrices are over $\mathbb{F}_3$.

Lemma 1. *If R is any matrix over a field with $RR^{\mathsf{T}} = 0$, then $Q = R^{\mathsf{T}}R$ is symmetric and $Q^2 = 0$. Its image is self-orthogonal. If also $Rj = 0$ and $j^{\mathsf{T}}j = 0$, then $\operatorname{im} Q + Fj$ is self-orthogonal.*

Proof. The matrix identities give symmetry and $Q^2 = R^{\mathsf{T}}(RR^{\mathsf{T}})R = 0$. Hence $\langle Qx, Qy\rangle = x^{\mathsf{T}}Q^2y = 0$. Moreover $Qj = R^{\mathsf{T}}Rj = 0$, so j is orthogonal to $\operatorname{im} Q$; its own norm is zero by hypothesis. □

For the fixed family, the exact integer scalar products of incidence rows are as follows. This table is checked for every pair of the 105 flags by the retained complete construction,

not inferred from a sampled orbit.

relation between the two flags	$(RR^{\mathsf{T}})_{f,g}$
$f = g$	24
same point or same line, but distinct	12
distinct points and lines, with one cross incidence	6
distinct points and lines, with neither cross incidence	3.

Two distinct points and distinct lines cannot have both cross incidences. Thus the table exhausts all cases and proves $RR^{\mathsf{T}} = 0$ over $\mathbb{F}_3$. Each row has sum 24, so $Rj = 0$, and the constant vector on 120 coordinates has norm zero. Set

$$K = \mathrm{im}(R^{\mathsf{T}}R), \qquad N = K + \mathbb{F}_3 j.$$

The exact row reductions give $\dim K = 15$ and $\dim N = 16$. Consequently the sum defining N is direct. Here the direct sum is an internal decomposition of modules on the same 120 coordinates, not a concatenation of codes on disjoint coordinate sets. Lemma 1 proves self-orthogonality of both codes. These ranks and the full table, including its underlying block lists, are obligations of the complete finite certificate.

2 Recognizing the socle

The literal permutations in `module_data.json` preserve the residual Steiner graph and generate a primitive permutation group $G \cong L_3(4)$ on $\mathcal{H}$, with point stabilizer $L_3(2)$. The certificate checks primitivity, the isomorphisms and stabilizer, so this is the specified degree-120 coset action in the source. It also checks invariance of K and N. The following elementary criterion makes the remaining module calculation transparent.

Lemma 2. *Let V be a finite-dimensional FG-module with an invariant filtration $0 = V_0 < V_1 < \cdots < V_s = V$, whose factors $D_i = V_i/V_{i-1}$ are simple. Let $N \subseteq V$ be a sum of simple submodules. If every G-homomorphism $D_i \to V$, for every i, has image in N, then $\mathrm{Soc}(V) = N$. A matrix representation on a d-dimensional space is absolutely irreducible if the algebra generated by its matrices is all $M_d(F)$.*

Proof. If $S \subseteq V$ is simple, choose the least i for which $S \subseteq V_i$. The intersections $S \cap V_j$ are submodules of S. Minimality implies $S \cap V_{i-1} = 0$, so projection embeds S into the simple factor D_i. It is therefore an isomorphism onto that factor. Its inverse, followed by $S \hookrightarrow V$, is a homomorphism $D_i \to V$; the hypothesis gives $S \subseteq N$. This holds for every simple submodule, proving $\mathrm{Soc}(V) \subseteq N$. The reverse inclusion follows from the hypothesis on N. For the last assertion, any invariant subspace is invariant under the whole matrix algebra. A nonzero vector can be sent to any chosen vector by an endomorphism, so that subspace is the whole space. The same argument holds after scalar extension because a basis for $M_d(F)$ remains a basis after extension. □

Here $V = \mathbb{F}_3^{120}$. The supplied basis certifies invariant prefixes of dimensions

$$0, 15, 34, 49, 64, 65, 84, 99, 100, 119, 120.$$

The ten factor dimensions and homomorphism-space dimensions are

i	1	2	3	4	5	6	7	8	9	10
$\dim D_i$	15	19	15	15	1	19	15	1	19	1
$\dim \mathrm{Hom}_G(D_i, V)$	1	0	0	0	1	0	1	1	0	1.

Triangularity of the generator matrices verifies invariance of all prefixes. For each factor the matrix algebra has dimension $(\dim D_i)^2$, proving simplicity by Lemma 2. The first prefix is exactly K. For each factor, the verifier forms *all* equations $XP_g = D_gX$ for the given generators and computes their full solution space. Every row of every basis solution belongs to N. Linearity then establishes the hypothesis for every homomorphism, not only a selected map. The constant line is simple and K is the first simple factor, so Lemma 2 yields

$$\mathrm{Soc}(\mathbb{F}_3^{120}) = N = K \oplus \mathbb{F}_3 j.$$

This calculation does not classify every irreducible module of $L_3(4)$. The source's classification assertion concerning three nonisomorphic faithful irreducible modules of dimension 15 is retained from its reference [13], Jansen–Lux–Parker–Wilson, *An Atlas of Brauer Characters* (1995). Our certificate identifies a particular such simple module; its faithfulness also follows because G is simple and the representation is nontrivial.

3 Weights and full permutation automorphisms

Complete enumeration in the certified bases gives

code	number of words	dimension	d	words of weight d	$d^\perp$
K	14348907	15	48	1260	4
N	43046721	16	48	1260	4.

The enumerator visits each of the 3^k coefficient vectors exactly once; independence of the basis makes these distinct words. It records the full weight distribution. The dual distributions are obtained from the classical MacWilliams identity

$$A_j^\perp = 3^{-k} \sum_{i=0}^{120} A_i \sum_{s=0}^{j} (-1)^s 2^{j-s} \binom{i}{s} \binom{120-i}{j-s},$$

with binomial coefficients outside their ranges taken as zero. In both cases the dual coefficients at weights $1, 2, 3$ vanish. At weight 4 they are 2100 for $K^\perp$ and 1260 for $N^\perp$. Thus the dual dimensions are 105 and 104, respectively, and the stated dual minimum distances follow. These are complete finite computations, not distance bounds inferred from self-orthogonality.

The orbit under G of the supplied weight 48 word contains 1260 distinct words and 630 distinct supports. Since $K \subset N$ and both codes have exactly 1260 minimum words, this orbit is the complete minimum-word set of both codes. Every coordinate-permutation automorphism of either code therefore acts on its support-incidence structure, with 120 coordinate vertices and 630 distinct support vertices.

Here is a short upper-bound certificate for the *full* automorphism group of that structure. Give the coordinate and support vertices two different initial colours. Refine colours by old colour and the multiset of neighbour colours. Successively individualize the coordinate vertices $1, 6, 48, 10, 2, 3$ in the literal certificate ordering. Their successive refined colour classes have sizes

$$120, 21, 2, 4, 2, 2.$$

After these choices all coordinate classes are singletons. Since the support neighbourhoods are distinct, an automorphism fixing the coordinates also fixes the supports. An automorphism fixing the previous choices must preserve every refined colour class. The successive orbit–stabilizer bounds therefore give order at most $120 \cdot 21 \cdot 2 \cdot 4 \cdot 2 \cdot 2 = 80640$.

The explicit permutations supplied in the certificate generate a group A of order 80640 and preserve each of K, N. Thus both code automorphism groups attain the upper bound and

equal A. The certificate checks $A' = G$ and supplies two commuting involutions generating a complement of order 4, disjoint from G. Consequently

$$\mathrm{Aut}(K) = \mathrm{Aut}(N) = A \cong L_3(4) : 2^2.$$

No graph-isomorphism program is needed for this upper bound. Such software was useful in discovering the permutations, but the retained proof checks literal permutations, ordinary group orders and the full refinement chain.

Limits. This note recovers only the stated socle and associated-code assertions of L050, Theorem 1(b)/Proposition 3. It does not establish the other source codes, minimum-dimension or global module-classification claims. It contains no new Lean formalization. The complete finite construction, module data, linear-algebra verifier and weight enumerator are required mathematical inputs and must be retained and charged in full when this proof is used. No numerical compression factor or historical priority claim is made.